SCHEMATIC DRAWING PACKAGE

abDraw

A Hands-On Tutorial

Capture · Route · Export
Multi-sheet schematics
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abDraw is a keyboard-friendly tool for drawing electronic schematics — blocks with named pins, buses, and auto-routed wires spread across as many sheets as your design needs. This guide walks you from an empty page to a complete, exportable datapath, in the order you'll actually build one. Work through it start to finish once; after that, the shortcut reference on the last page is all you'll need for day-to-day use.

CONTENTS

1 The workspace

When abDraw opens you're looking at four regions. Learn their names now; the rest of the guide uses them constantly.

- **Toolbar** (top) — the row of tool buttons. The leftmost is **Select**; the rest create the common shapes, wires, primitives, connectors and text. Hover any button for a tooltip. Style controls (Outline color, Fill, Line Style, Width) live here too. Less-used shapes — Square, Triangle, Mux, Adder, the Note Line/Arrow, and the block arrows — live in the **Shapes** menu.
- **Canvas** (center) — the drawing surface. The white rectangle is the current **page**; the gray area around it is the pasteboard. Anything off the white page won't print.

- **Sheet tabs** (below the canvas) — one tab per sheet in the package. Click to switch, double-click to rename, use the add/delete controls to manage them.
- **Status bar** (bottom) — live feedback: the active tool, hints for the current gesture, and the zoom percentage.

> MENTAL MODEL
>
> Everything you draw is a **shape** living at a real position on the page. Wires are shapes that *bind to a specific named pin* on a block — that binding is what keeps them attached when you move things around. Keep that in mind and the router's behavior later will feel obvious.

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2 Grid, snap, zoom & pan

A tidy schematic is a snapped schematic. Two toggles in the **View** menu govern this:

- **Show Grid** `Ctrl+G` — the dotted alignment grid. On by default.
- **Snap to Grid** `Ctrl+H` — forces new shapes and drags onto grid points. Leave this *on* while you build; wires only stay clean if their endpoints land on the grid.
- **Background** — **View › Background** toggles a light or dark canvas. Dark mode re-inks only the near-black elements (outlines, wires, pins, text) to a light tone; colors you chose on purpose — blue nets, amber, a custom fill — stay exactly as set. The choice is saved with the file and defaults to Light.

Zoom works two ways:

- **Mouse wheel** — zooms toward the pointer, so the feature under your cursor stays put. This is the fast, everyday zoom: roll out for the big picture, roll in to place a pin.
- The **View** menu — **Zoom In** `Ctrl +`, **Zoom Out** `Ctrl -`, **Zoom to Fit** (whole page in view), and **Actual Size** `Ctrl 0` (back to 100%).
- **Pan** with the scrollbars at the canvas edges.

> GOOD TO KNOW
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> Zoom is purely a *view*. It never changes what's saved or exported — a schematic exported at 300% looks identical to one exported at 100%. Zoom freely; you can't hurt anything.

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3 Your first block

Most functional blocks in a datapath are rectangles with a name and some pins. Let's make one.

1. Click the **Rectangle** tool in the toolbar.
2. On the canvas, press and drag to sweep out a box, then release. The block appears, snapped to the grid.
3. Switch to the **Select** tool. Click the block's *border* to select it (dashed blue outline + resize handles appear).
4. Give it a name: **Edit › Add Label to Shape** `Ctrl+L`. Type e.g. `ALU` and confirm. The label dialog offers font, size, bold, italic and color, and accepts multiple lines — press `Enter` for a new line and `Ctrl+Enter` to confirm.
5. Drag the block to reposition, or drag a corner handle to resize. Connected wires follow.

> SELECTING HOLLOW SHAPES
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> An unfilled rectangle or square is selected by clicking its **outline**, not its empty interior. If a click in the middle seems to miss, aim for the border. (Filled shapes are clickable anywhere.)

To fill a block, select it and use the toolbar **Fill** button — *No Fill* or *Choose Color...*. Your choice also becomes the default for the next shape. Interior text (pin names, the adder +) automatically flips to white or black for contrast against the fill. The toolbar **● Outline** swatch sets the outline (border, lines and label) color the same way — and when you open **Fill** or **Outline** with a shape selected, the picker opens on that shape's current color so you can nudge it. To copy one shape's look onto others, select the target(s) and use **Edit › Match Selected Color** `Ctrl+M`.

Rectangles and squares can have **rounded corners**: with one selected (or none, to set the default), choose **Shapes › Rounded Corners...** and enter a radius in pixels (0 = square). The radius rides along on resize, saves with the file, and exports rounded.

4 Pins & ports

Pins are the named connection points on a block's edges. They're what wires attach to, and what the netlist reads. Select a block and open **Edit › Edit Pins...** `Ctrl+P`.

In the Edit Pins dialog you can:

- Add a pin, choosing its **side** (Left / Right / Top / Bottom) and name.
- Reorder pins on a side — they stay evenly spaced and grid-aligned.
- Use **Add Range...** for buses: give a prefix, start, end, and zero-pad width and abDraw generates `D00…D15` in one step (up to 256 at a time; existing names are skipped and reported).

For clock and power-style pins, **Edit › Special Pins...** `Ctrl+Shift+P` handles the special cases. Once pins exist, they move, resize and rotate with the block, and any wire bound to a pin tracks it automatically.

5 Wiring & the router

Wires are where abDraw earns its keep. You have three ways to draw them, and one rule that ties them together.

The three wire tools

- **Line / Arrow** — a straight wire. Press on the source pin, drag to the target pin, release. The endpoints snap to the nearest pins and bind.
- **Ortho Line / Ortho Arrow** — a wire with 90° turns. **Left-click** to drop each corner, **right-click** or `Enter` to finish, `Esc` to cancel. This is your main tool for datapath wiring.

THE ONE RULE

Any wire with a pinned end is **owned by the auto-router**. When you move a connected block, the wire rebuilds itself so it always leaves each pin *perpendicular to that pin's edge*, and fans of wires sharing an edge stagger into separate channels instead of overlapping. You draw intent; the router keeps it clean as things move.

When you want to route by hand

Select a wire to reveal its blue **waypoint handles**. Drag a handle to bend the wire; **double-click** the wire to insert a new waypoint; drag the orange mid-segment handle to slide a segment sideways. A wire you've hand-routed is honored as drawn — the router won't override it — until you move a connected block or choose **Edit › Auto-Route Wire** to hand it back to the router.

Arrowheads: **Edit › Wire Arrowheads** sets none / one / both ends, or cycle them with `Ctrl+Shift+A`.

NOTES VS. WIRES

The **Note Line** and **Note Arrow** tools look like wires but are purely annotation — they never bind to a pin, never appear in the netlist, and never trigger junctions. Use them for callouts and comments. Give a Note Line a dash pattern via the toolbar **Line Style...** button. Their endpoints snap to a nearby shape's edge (radially on an oval, to the border on a box) so a callout lands cleanly on the block it points at — without any electrical binding.

Block arrows for diagrams

For dataflow and block-diagram callouts, the **Shapes** menu offers three fat **block arrows**: **Block Arrow** (filled), **Block Arrow (outline)**, and **Block Arrow (double)** with a head at each end. Drag to draw at any angle. Like notes they're non-electrical — no pins, no netlist. Tune them with **Shapes › Block Arrow Thickness...** (body width in px), set the interior via the toolbar **Fill**, and rotate a selected one finely with `R` / `Shift+R` (±15°) or an exact angle via **Shapes › Rotate Block Arrow...**.

6 Buses & bit-slices

A multi-bit connection is drawn as a single thicker **bus**. Select a wire and toggle it with Edit › Toggle Bus `Ctrl+B`.

To show which bits a wire carries, add a **bit-slice label** via Edit › Bit-Slice Label… — e.g. `[7:0]`. If the label sits awkwardly, nudge it with **Bit-Slice Label Distance…**, or just drag it; its position is remembered and exports where you place it. Where three or more wire ends meet, abDraw drops a **junction dot** automatically; a wire ending on the middle of another creates a T-junction dot too.

7 Primitives

Beyond plain shapes, abDraw draws the components a CPU datapath is made of, each with its pins pre-placed:

- Mux — drag a box, then enter the number of inputs. It draws as a trapezoid with binary-labeled inputs, a `sel` pin, and a `y` output. The top/bottom pins ride the slanted edge correctly.
- Register — a flip-flop with `D`, `Q`, and a clock pin marked by the usual edge triangle.
- Adder — a block with a + glyph.
- Off-Page / On-Page connectors — covered next.

8 Net labels & connectors

You don't have to physically wire everything. abDraw connects by **name** as well as by line:

- **Net labels** — select a wire and give it a name via Edit › Net Label… `Ctrl+Shift+N`. Two wires with the *same* net name are the same electrical net — no line needed between them.
- **On-Page connector** — same name links nodes on *this* sheet.
- **Off-Page connector** — same name links nodes *across* sheets, so a bus can leave sheet 1 and arrive on sheet 3. Rotate its attach side with Rotate Connector Port `Ctrl+R`, and rename via **Rename Connector…**.

SEE A NET AT A GLANCE

Select a net-labeled wire or a connector and every other wire/connector on the same net lights up with a blue halo. It's the quickest way to confirm that two things you *think* are connected actually share a name.

9 Selecting, groups & arrange

- **Single select** — click a shape (its border, if hollow) with the Select tool.
- **Marquee select** — with the Select tool, drag a rubber-band box across empty canvas. Every shape it touches is selected and outlined in green. Works at any zoom.
- **Select All** — `Ctrl+A`.
- **Group move** — drag any member of a green group and the whole group moves together. A wire pinned to a block *outside* the group stays attached to that block; wires between in-group blocks ride along.
- **Delete** — `Delete` removes the selection (single or group) in one undo step.

Copy / paste with `Ctrl+C` / `Ctrl+V`; each paste offsets a little so copies don't stack, and **Paste in Place** `Ctrl+Shift+V` drops it exactly where the original sits. Overlapping symbols? Use **Arrange › Bring to Front** / **Send to Back** — the whole symbol (shape, label and pins) restacks as a unit, and the z-order is saved with the file.

10 Multi-sheet packages

A schematic is a **package** of one or more sheets. Use the sheet tabs to add, rename, delete, **duplicate** and switch sheets — **Duplicate** deep-copies the active sheet (all shapes plus its page size), drops the copy in right after it, and makes it active, so you can spin a variant off an existing sheet without redrawing. Each sheet has its own page size — Letter, Legal, Tabloid, ANSI, an A-series size, or Custom — and its own title block showing "Sheet N of M" and the date. Off-page connectors (section 8) are how signals travel between sheets: same name, same node. Save the whole package to one file; it reopens with every sheet, page size and title block intact.

11 Netlist & validation

When the drawing is done, let abDraw check your work. Both commands live in the **Schematic** menu:

- **Validate Schematic** — rings every *unconnected* pin in red so you can spot the wire you forgot. Run it often.
- **Generate Netlist…** — writes a `.net` file listing every electrical net, resolved across all sheets (wires, net labels, on- and off-page connectors all counted together).

12 Saving & exporting

Save `Ctrl+S` writes the entire package — all sheets, pins, routing and z-order — to one file you can reopen later. That's your working format.

To share or print, use **File › Export**:

- **Sheet as PNG** / **All Sheets as PNG** — raster images.
- **Sheet as PDF** — the current sheet.
- **Package as PDF (multi-page)** — every sheet, one page each, at its own page size.

> PRINT-FRIENDLY TEXT
>
> On export, label text is rendered in solid black so faint on-screen colors (like the light-blue net labels) stay crisp on paper. Pin and connector names on dark-filled blocks keep their contrast color so they remain legible. What you place is what prints.

13 Worked example: an ALU stage

Let's tie it together by capturing one stage of the abCore16 datapath — two source registers feeding an ALU through an input mux, with the result latched back. Follow along; every step uses something from the sections above.

1. **Place the registers.** Register tool → drag two blocks on the left, one above the other. Label them `RA` and `RB` `Ctrl+L`.
2. **Place the mux.** Mux tool → drag a box to the right of the registers, enter `2` inputs. Its inputs line up with `RA.Q` and `RB.Q`.
3. **Place the ALU.** Rectangle tool → a tall block on the right; label it `ALU`. Open Edit Pins `Ctrl+P` and add `A` and `B` on the left, `Y` on the right, and an `op` pin on top. For a wide bus, use **Add Range…** to make `Y00…Y15` at once.
4. **Wire it up.** Ortho Line tool → from `RA.Q` click across to the mux input, right-click to finish. Repeat for `RB.Q`, then mux `y` → `ALU.A`. Notice each wire leaves its pin perpendicular and the two register outputs stagger into separate channels.
5. **Make it a bus.** Select the `Y` output wire, `Ctrl+B` to make it a bus, then **Bit-Slice Label…** → `[15:0]`.
6. **Send the result off-page.** Off-Page connector → place it at the end of the `Y` bus, name it `RESULT`. On another sheet, a second `RESULT` connector picks the bus back up.
7. **Tidy & verify.** Marquee-select the whole stage and nudge it into place. Run Schematic › Validate Schematic — fix any red-ringed pins. Then **Generate Netlist…** to confirm `RA.Q`, the mux, and `ALU.A` land on one net.
8. **Deliver.** `Ctrl+S` to save the package, then **File › Export › Package as PDF**.

That's the whole loop — place, pin, route, name, verify, export. Every larger schematic is just more of the same.

14 Shortcut reference

FILE

New	Ctrl+N
Open	Ctrl+O
Save	Ctrl+S
Save As	Ctrl+Shift+S

EDIT

Undo / Redo	Ctrl+Z / Ctrl+Y
Copy / Paste	Ctrl+C / Ctrl+V
Paste in Place	Ctrl+Shift+V
Delete	Delete
Select All	Ctrl+A
Add Label	Ctrl+L
Match Selected Color	Ctrl+M

PINS & WIRES

Edit Pins	Ctrl+P
Special Pins	Ctrl+Shift+P
Toggle Bus	Ctrl+B
Cycle Arrowheads	Ctrl+Shift+A
Net Label	Ctrl+Shift+N
Rotate Connector	Ctrl+R
Finish ortho wire	Enter / R-click
Cancel ortho wire	Esc
Rotate block arrow	R / Shift+R

VIEW

Show / Snap Grid	Ctrl+G / Ctrl+H
Zoom In / Out	Ctrl+= / Ctrl+-
Actual Size	Ctrl+0
Zoom to cursor	Mouse wheel